



FOUR-COURSE SYSTEM

## COURSE 3 - DATA ENGINEERING 2026

# C3-014C Gate Review - Release Configuration Snapshot

Protected competency diagnosis | no copy-ready final solution

|                          |                                                                                                        |
|--------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Artifact</b>          | C3_014C_GATE_REVIEW_PROTECTED_RC1                                                                      |
| <b>Status</b>            | PROTECTED - OPEN AFTER FROZEN ATTEMPT                                                                  |
| <b>Teaching standard</b> | M01 beginner-first teacher loop; book remains sufficient without video.                                |
| <b>Bridge boundary</b>   | Course 2B competencies may be assumed; all new engineering concepts still start from first principles. |
| <b>Brand</b>             | Charcoal #1F2937   Teal #14B8A6   Soft Blue #3B82F6   AMOLED #000                                      |

House rule: understand -> follow once -> see expected result -> break/fix -> do a changed version -> validate -> explain -> save evidence. A running command alone never counts as mastery.

## Use this only after the Gate attempt is frozen

For each competency, mark PASS only from evidence. A failure in one competency blocks Gate PASS regardless of strength elsewhere. Do not convert this into a numeric average.

### DIAGNOSIS TABLE

| ID  | Look for                                                                                           | If weak, repair only this                                                                                                      |
|-----|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| C1  | Input/output grain, rules, failure, rerun and independent validation are explicit.                 | Restate one-row grain, required fields, output contract, failure, rerun and independent control on changed data.               |
| C2  | Correct shell/path; inspect before destructive action; source/evidence handling safe.              | Reproduce smallest filesystem operation in Linux-native test folder; use inspection-first recovery.                            |
| C3  | Reviewable branch/commits; no secret/venv/bytecode; diff understood.                               | Use status/diff/staged-diff; remove disposable/secret artifacts; create a fresh changed branch/commit exercise.                |
| C4  | Clean venv/dependency recreation proven.                                                           | Delete only disposable venv, recreate from dependency files, rerun tests.                                                      |
| C5  | Core/CLI boundary + documented config precedence + no secret leakage.                              | Separate core/CLI; prove config precedence; run secret scan.                                                                   |
| C6  | Actionable logs; known failure non-zero; unexpected error not swallowed.                           | Reproduce one known validation failure + one unexpected failure; ensure non-zero/trace context.                                |
| C7  | Tests protect contract incl. failure/edge case; deliberate regression is catchable.                | Add a test that fails under an intentionally injected regression, then restore code.                                           |
| C8  | Useful help, stable invocation/output and machine-readable failure behavior.                       | Repair help/argument/exit/output contract on a changed tiny CLI.                                                               |
| C9  | Image/container/mount/runtime reasoning correct; real Docker evidence or honest approved fallback. | Explain image/container/mount; if real Docker available prove client+server and tiny run; otherwise document blocker honestly. |
| C10 | Safe rerun demonstrated; operator can reproduce/validate/recover from runbook.                     | Define intended rerun state, run twice on changed case, compare, and revise runbook.                                           |

### CRITICAL FAILURE CHECK

- Real secret/token/key exposed in submitted source/log/evidence.
- Fabricated command/runtime/Docker evidence.
- Destructive unsafe operation or data loss without justified recovery control.
- Contract failure exits success or publishes a known-invalid result.
- Protected review opened/copied before first attempt is frozen.
- First-attempt evidence overwritten/rewritten after review.
- Unresolved plausible wrong result with no independent validation.

### NEXT ROUTE

If Gate A has targeted weaknesses, complete only assigned remediation + changed micro-retest. Gate B/C do not unlock automatically. A fresh full retest requires explicit assignment and must use its unseen case.